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ORIGINAL PAPER

Nail in the Coffin or Lifeline? Evaluating the Electoral Impact of COVID-19 on President Trump in the 2020 Election

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Abstract
From the onset of the first confirmed case of COVID-19 in January 2020 to Election Day in November, the United States experienced over 9,400,000 cases and 232,000 deaths. This crisis largely defined the campaign between former Vice President Joe Biden and President Donald Trump, centering on the Trump administration′s efforts in mitigating the number of cases and deaths. While conventional wisdom suggested that Trump and his party would lose support due to the severity of COVID-19 across the country, such an effect is hotly debated empirically and theoretically. In this research, we evaluate the extent to which the severity of the COVID-19 pandemic influenced support for President Trump in the 2020 election. Across differing modeling strategies and a variety of data sources, we find evidence that President Trump gained support in counties with higher COVID-19 deaths. We provide an explanation for this finding by showing that voters concerned about the economic impacts of pandemic-related restrictions on activity were more likely to support Trump and that local COVID-19 severity was predictive of these economic concerns. While COVID-19 likely contributed to Trump's loss in 2020, our analysis demonstrates that he gained support among voters in localities worst affected by the pandemic.

Keywords COVID-19 pandemic · 2020 US presidential election · Donald Trump · Rally-'round-the-flag · Retrospective voting

COVID-19 in the Context of the 2020 U.S. Election

Conventional wisdom holds that the COVID-19 pandemic directly and seriously damaged Donald Trump's reelection effort in 2020. The depth of the crisis during 2020 is hard to overstate: from the onset of the first confirmed case of

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the coronavirus (COVID-19) in January 2020 to Election Day in November, the United States experienced over 9,400,000 cases and 232,000 deaths. News articles frequently described the Trump administration's handling of the pandemic as politically disastrous.[1],[2] Political scientists commenting contemporaneously on the election often echoed similar statements.[3],[4] Even Trump's own aides told *The Washington Post* in an election post-mortem that Trump's mismanagement of the virus was the primary cause of his loss to Biden.[5] These beliefs were understandable, given that public opinion surveys showed that a majority of Americans disapproved of Trump's handling of the pandemic.[6] There is also little doubt that COVID-19 loomed large in the public consciousness, as reflected by public opinion surveys,[7] high levels of (often politicized) media coverage (Hart et al., 2020; Pearman et al., 2021), and a daily death toll equivalent to the September 11, 2001 attacks (Woolf et al., 2021).

Despite the 2020 consensus that COVID-19 directly hurt Trump's reelection bid, subsequent investigations have yielded mixed results. A study by Baccini et al. (2021) finds that county-level COVID-19 rates were negatively associated with Trump's vote share. Mendoza Aviña and Sevi (2021) make similar findings at the individual level, concluding that voters who knew someone who contracted COVID-19 were less likely to vote for Trump (with a greater effect among those who reported knowing someone who died as a result). By contrast, other investigations found that Trump actually performed better in states[8] and counties[9] with higher COVID-19 mortality rates.

This empirical disagreement over the effects of COVID-19 on the 2020 election mirrors conflicting theoretical expectations that can be drawn from political science

[1] Los Angeles Times: 'They've lost the narrative:' Trump's virus stance backfiring politically (10/6/2020). https://www.latimes.com/politics/story/2020-10-06/trump-campaign-struggles-with-coronavirus-message

[2] Politico: 'This is the worst nightmare for the Trump campaign' (10/2/2020). https://www.politico.com/news/2020/10/02/the-campaign-as-we-knew-it-is-over-trumps-covid-test-upends-the-race-425020

[3] The Washington Post: The coronavirus is hurting Trump's 2020 campaign (5/6/2020).https://www.washingtonpost.com/outlook/2020/05/06/coronavirus-is-killing-trumps-2020-campaign/

[4] Los Angeles Times: Pandemic? What pandemic? Trump reelection ads ignore coronavirus (6/19/2020).https://www.latimes.com/politics/story/2020-07-19/trump-coronavirus-presidential-campaign-advertising-battlegrounds

[5] The Washington Post: "How Trump's erratic behavior and failure on coronavirus doomed his reelection (11/7/2020).https://www.washingtonpost.com/elections/interactive/2020/trump-pandemic-coronavirus-election/

[6] Pew Research: Majority of Americans disapprove of Trump's COVID-19 messaging, though large partisan gaps persist (9/15/2020).https://www.pewresearch.org/fact-tank/2020/09/15/majority-of-americans-disapprove-of-trumps-covid-19-messaging-though-large-partisan-gaps-persist/

[7] CNBC: Here's what mattered most to voters in the 2020 election, according to exit polls (11/3/2020).https://www.cnbc.com/2020/11/03/exit-polls-heres-what-mattered-most-to-voters-in-the-2020-election.html

[8] FiveThirtyEight: How Much Did COVID-19 Affect The 2020 Election? (1/27/2021)https://fivethirtyeight.com/features/how-much-did-covid-19-affect-the-2020-election/

[9] NPR: Many Places Hard Hit By COVID-19 Leaned More Toward Trump In 2020 Than 2016 (11/6/2020)https://www.npr.org/sections/health-shots/2020/11/06/930897912/many-places-hard-hit-by-covid-19-leaned-more-toward-trump-in-2020-than-2016

literature on the effects of crises on leader approval. The classic retrospective accountability framework stipulates that the public punishes leaders who mishandle crises (Ashworth, 2012; Healy & Malhotra, 2009, 2013; Fiorina, 1978; Ferejohn, 1986). Under this model, we would reasonably expect that Trump paid an electoral price for his perceived mishandling of the pandemic. Another possible framework is the “rally ’round the flag” effect, wherein the public politically supports leaders in times of crisis – with larger effects for worse crises (Mueller, 1970; Lee, 1977; Oneal & Bryan, 1995; Baum, 2002). If the latter framework dominates, we should expect that Trump benefited electorally from the COVID-19 crisis, and the benefits should be stronger in areas hardest hit by the pandemic. Both the rally and retrospective accountability frameworks have strong and ongoing empirical support in studies of American politics.

In order to address this empirical and theoretical uncertainty as it pertains to the effects of COVID-19 in the 2020 presidential election, we propose and carry out a novel series of tests. We ultimately find that higher COVID-19 rates in a given county were associated with higher levels of Trump support. We also show that individual-level concern about the economic impacts of government-imposed COVID-19 restrictions was correlated with increased Trump support. Both of these findings contrast with conventional wisdom, and highlight the need for caution when making political prognostications about the electoral effects of national crises. We further show that voters in localities worst hit by the pandemic were more likely to be concerned about the economic impacts of COVID-19 restrictions—yielding an electoral boost for Trump in these areas. This finding dovetails with anecdotal observations that the Trump campaign sought to emphasize the economic costs and minimize the public health dimension of COVID-19 restrictions during the 2020 campaign. While this strategy was ultimately unsuccessful, given that nearly 60% of voters were more concerned that COVID-19-related restrictions would be lifted too quickly (rather than not quickly enough), we show that Trump’s economic appeals on COVID-19 were surprisingly effective in localities hit hardest by the pandemic. This finding suggests that while Trump did benefit electorally in areas worst affected by the pandemic, the mechanism was not a rally effect. Instead, the intensity of the crisis in these localities increased the salience of voters’ pandemic-related economic concerns—concerns which Trump emphasized in his 2020 campaign.

Contrasting Frameworks: Retrospective Accountability vs. The Presidential Rally

A central debate in American politics revolves around whether voters reward or punish elected officials during times of national crisis. Popularized by Key (1966), the retrospective voting theory suggests that voters rationally cast votes based on the past performance of incumbent politicians in office (rather than stated policy positions, for instance). There is considerable evidence that voters consider their recent economic circumstances when casting a ballot (e.g., Fiorina, 1978; Kramer, 1971; Achen & Bartels, 2016). In the context of national crises, Healy and Malhotra

(2009) found that voters reward the incumbent party for spending on disaster relief (though, perversely, not for disaster prevention).

Under the retrospective voting model, we expect Trump to be electorally damaged by the COVID-19 pandemic. These sentiments rippled across the United States—agency directors, political opponents, and medical professionals alike criticized the President's response to the COVID-19 pandemic. Indeed, during a presidential debate, Joe Biden directly linked the pandemic to Trump's ability to govern, stating "anyone that's responsible for that many deaths should not remain president of the United States of America." Warshaw et al. (2020) find that in the summer prior to the 2020 election it appeared that voters in states with high incidences of COVID-19 deaths reduced their intention to support President Trump and his congressional Republican allies. While conducting this research on voting intention over the summer of 2020 rather than on observed Election Day results, Warshaw et al. (2020) conclude with the expectation that voters will penalize President Trump and his Republican allies for pronounced COVID-19 casualties. In the final weeks leading up to the election, nearly 60% of Americans disapproved of President Trump's handling of the pandemic.[10]

A natural rejoinder to the expectation of retrospective voting can be found in Achen & Bartels's 2016 famous critique of retrospective voting (and the broader "folk theory of democracy"). They find that voters irrationally punish incumbents for incidents outside of their control (e.g., natural disasters). In one particularly illuminating example, Achen and Bartels (2016) find that voters in the 1918 congressional election punished incumbents regardless of their policy response to the Spanish Flu. In other words, while voters may not reward politicians who respond "well" to a disaster like a pandemic (contra Healy & Malhotra, 2009), per Achen and Bartels (2016) they do at least punish incumbents who preside over crises (regardless of their response). Though this critique does substantially limit the credibility of retrospective voting, it does not change the direction of our expectations that a retrospective electorate would punish Trump for presiding over a major disaster.

The increasing salience of partisanship in American elections further complicates the narrative that voters would retrospectively punish Trump in 2020. In a political landscape where voters increasingly rely on partisan cues at the ballot box (e.g., Jacobson, 2015; Smidt, 2017), increasingly dislike out-partisan politicians (e.g., Jacobson, 2017), and adopt ideological positions from party leaders (Lenz, 2013), it is unsurprising that studies show that voters' retrospective evaluations are not immune from partisan bias (e.g., Heersink et al., 2020; Achen & Bartels, 2016). Further, research has found that voters are more likely to favorably assess economic conditions (Gerber & Huber, 2010) and are substantially less likely to blame the incumbent government (Bisgaard, 2015) for an economic catastrophe when they know that their party is in power. In short, while COVID-19-based retrospective voting may have hurt Trump in 2020, partisan motivated-reasoning may have dampened such an effect.

[10] Pew Research Center: Majority of Americans disapprove of Trump's COVID-19 messaging, though large partisan gaps persist (10/15/2020)https://www.pewresearch.org/fact-tank/2020/09/15/majority-of-americans-disapprove-of-trumps-covid-19-messaging-though-large-partisan-gaps-persist/

An alternative model of political behavior—the rally effect—makes contrary predictions of how the public responds to a crisis (Baum, 2002; Mueller, 1970). Baum (2002) observed that the U.S. president enjoys a surge in their approval ratings during times of war and in the wake of international crises.[11] Today, there is ample empirical evidence in support of rally effects in the United States (Baker & Oneal, 2001; Chapman & Reiter, 2004; Chatagnier, 2012).

Like war, the COVID-19 pandemic can be likened to a national crisis with potential unifying effects (Schraff, 2020; Warshaw et al., 2020). The onset of COVID-19 was an exogenous shock that was dramatic in nature, international, and directly involved the president who set the agenda on a public health response. This war analogy was even explicitly echoed by Trump, who declared himself a "wartime president" in the battle against COVID-19 and by Republican Senator John Cornyn (R-Texas) who defended pandemic fiscal spending by arguing that "this is really a wartime footing" (Oprysko & Luthi, 2020). As the logic follows, as the number of fatalities increase, citizens should champion, not punish, national leaders in the face of a crisis. As seen in studies of public opinion during conflict, rally effects tend be larger as the number of fatalities increase (Erikson, 1990, 2016) since more dramatic events are more likely to prompt a unified response (Chowanietz, 2011). Put simply, the rally-round-the-flag effect produces somewhat somber predictions about the pandemic: As fatalities increase, we should anticipate more unity around political leaders.

Along these lines, Giommoni and Loumeau (2020) report that city council incumbents running for reelection in French municipalities under stricter lockdowns experienced higher vote shares than incumbents in municipalities under softer lockdowns. Indeed, there is much comparative support for COVID-19 pandemic-induced rally effects with respect to public approval for national leaders, trust in government and political institutions, and electoral support for incumbents and the party of the national executive (e.g., Devine et al., 2020; Yam et al., 2020; Giommoni & Loumeau, 2020; Bol et al., 2020; Esaiasson et al., 2020; Schraff, 2020; Devine et al., 2020). Instead of castigating their elected officeholders for their perceived role in the pandemic-induced public health crises and economic downturns, recent literature reveals that public support for officeholders increased when lockdowns began, death tolls spiked, and economies tanked (Jennings, 2020; Esaiasson et al., 2020; Devine et al., 2020; Bol et al., 2020).

[11] For example, a Gallup poll immediately following the September 11th terrorist attacks showed a marked increase in President Bush's approval rating: From 51% to 86%. In order for these rally effect to occur, Mueller (1970) theorized that the crisis must meet three criteria: (1) "Be international, [...] (2) involve the United States and the President directly, (3) [and] be specific, dramatic, and sharply focused" (Mueller, 1970, p. 21).

Trump, the Economy, and COVID-19

The retrospective accountability and rally theories provide two plausible and competing expectations about the effect of COVID-19 on Trump's reelection efforts in 2020. Existing studies are similarly conflicting, with disagreement about both the direction and magnitude of such an effect. In the retrospective accountability camp, Mendoza Aviña and Sevi (2021) find that COVID-19 was negatively associated with voting for Trump. Baccini et al. (2021) come to the same conclusion at the aggregate level, showing that Trump's 2020 vote share and county COVID-19 case rates were negatively associated.

By contrast, other studies have found evidence in line with the expectations of a rally hypothesis. Numerous studies identified an initial rally in support for Trump at the beginning of the pandemic in spring 2020—including among Democratic voters (Yam et al., 2020; Dunn, 2020; Dzhanova, 2020). According to analysis presented in a study reported in *NPR*,[12] election support for Trump increased in counties worst hit by COVID-19.

To reconcile these conflicting findings and conflicting theories, we argue that concerns about the economic impacts of COVID-19 policies and Trump's appeal to voters on those grounds complicate a straightforward application of either the retrospective voting or rally models. Academic studies and data journalistic examinations that find voter-level backlash to Trump over his handling of COVID-19 have tended to focus on perceptions related to the personal or public health risks of COVID-19. However, as Neundorf and Pardos-Prado (2021) point out, public concerns about COVID-19 are multidimensional. Voters are concerned not only about public health policy, but also to economic costs associated with the pandemic and policy responses to it. Studies of the electorate in 2020 also reveal a partisan divide in how voters assess COVID-19 risks, with Republicans expressing greater concern about the economic effects of restrictions and Democrats exhibiting greater preference for social distancing and lockdowns (Craig et al., 2022; de Bruin et al., 2020).

Not only were economic concerns about COVID-19 restrictions an important dimension of COVID-19 attitudes, but the 2020 Trump campaign viewed those concerns to be an electoral asset—especially among the Republican base. As early as the spring of 2020, Trump tweeted that "WE CANNOT LET THE CURE BE WORSE THAN THE PROBLEM ITSELF"[13] and in response to state-imposed stay-at-home orders "LIBERATE MINNESOTA!," "LIBERATE MICHIGAN!," and "LIBERATE VIRGINIA."[14] During a town hall, President Trump commented that "you're going to lose more people by putting a country into a massive recession or depression," suggesting that a Democratic victory would mean more economic

[12] NPR: Many Places Hard Hit By COVID-19 Leaned More Toward Trump In 2020 Than 2016 (11/6/2020)https://www.npr.org/sections/health-shots/2020/11/06/930897912/many-places-hard-hit-by-covid-19-leaned-more-toward-trump-in-2020-than-2016

[13] The New York Times: Trump Says Coronavirus Cure Cannot 'Be Worse Than the Problem Itself' (3/23/2020)https://www.nytimes.com/2020/03/23/us/politics/trump-coronavirus-restrictions.html

[14] The Detroit News: Trump tweets 'liberate' Michigan, two other states with Dem governors (4/17/2020)https://www.detroitnews.com/story/news/politics/2020/04/17/trump-tweets-liberate-michigan-other-states-democratic-governors/5152037002/

shutdowns and more prolonged restrictions that would close large portions of the American economy.[15] Other GOP officials were quick to align with Trump's messaging, with Texas Lieutenant Governor Dan Patrick suggesting that grandparents should be willing to risk death in order to end restrictions in the state[16] and Florida Governor Ron DeSantis saying "The cost of the lockdowns are born by working-class people. The benefit is to the Zoom-class, the upper-income people who can work from home. Not everyone can do that. You have to go out."[17] In the week leading up to the election, Trump's COVID-19 messaging was firmly focused on the economic cost of restrictions, as reflected in Tweets like "Biden wants to LOCKDOWN our Country, maybe for years. Crazy! There will be NO LOCKDOWNS. The great American Comeback is underway!!!" and "The Biden Lockdown will mean no school, no graduations, no weddings, no Thanksgiving, no Christmas, no Fourth of July."[18]

The preference among Republican elites for economic reopening over public health precautions was reflected in state governments, where GOP governors were less likely to impose public health measures (e.g., stay-at-home orders) than Democratic governors (Baccini & Brodeur, 2021; Shay, 2020). There is ample evidence that these attitudes percolated to Republican voters as well, who were less likely to modify their behavior in response to the pandemic (e.g., by masking or staying at home) (Gadarian et al., 2021; Clinton et al., 2021; Kassas & Nayga, 2021; Allcott et al., 2020; Hsiehchen et al., 2020), less likely to comply with state shutdown orders (Camobreco & He, 2022), and were more likely to be concerned about the economic impacts of restrictions than on public health and worker safety risks associated with reopening (Green et al., 2020; Craig et al., 2022; de Bruin et al., 2020).

In this paper, we will examine the relationship between COVID-19 attitudes, vote choice, and local COVID-19 case rates and death rates at the aggregate and individual levels. At the aggregate level, we aim to answer whether the local experience of the disaster led to a retrospective electoral penalty or to a rally for Trump. At the individual level we explore how attitudes about COVID-19 and public health affect vote choice. In addition, we evaluate whether concerns about economic restrictions also affected vote choice, and if so in what direction. At the individual level, we expect to find that economic concerns about COVID-19 electorally assisted Trump while concerns about public health hurt him. Such a finding would help provide an explanation for the mixed (and seemingly contradictory) findings on the effect of COVID-19 on the 2020 presidential election.

15 The New York Times: Trump's Baseless Claim That a Recession Would Be Deadlier Than the Coronavirus (5/6/2020)https://www.nytimes.com/2020/03/26/us/politics/fact-check-trump-coronavirus-recession.html

16 The Texas Tribune: Dan Patrick says "there are more important things than living and that's saving this country" (4/21/22)https://www.texastribune.org/2020/04/21/texas-dan-patrick-economy-coronavirus/

17 The Palm Beach Post: DeSantis dismisses lockdowns, mask mandates as coronavirus cases spread (11/30/2020)https://www.palmbeachpost.com/story/news/2020/11/30/gov-ron-desantis-mocks-lockdowns-mask-mandates-coronavirus-cases-spread/6467204002/

18 The Guardian: Donald Trump tries to stoke fears of Covid lockdown under Joe Biden (11/2/2020)https://www.theguardian.com/us-news/2020/nov/02/trump-biden-coronavirus-covid-lockdown

In addition, we expect that other contextual factors may have conditioned any individual-level relationship—potentially providing an explanation for conflicting findings[19] in previous analyses of the effect of COVID-19 on 2020 election. We further anticipate that the magnitude of the effects of both COVID-19 public health concerns and economic restriction concerns on presidential vote choice were conditioned by partisanship, given the Trump campaign's emphasis of the economic dimension and the Biden campaign's emphasis of the public health dimension (Green et al., 2020). Finally, we will examine whether the magnitude of the individual-level relationships between COVID-19 concerns, both public health and economic, and support for Trump may have been conditioned by aggregate-level COVID-19 conditions (e.g., county-level death rates and case rates).

Research Design

Specifying the Aggregate Model of Electoral Accountability

To evaluate whether voters electorally punished President Trump for his role in the severity of the pandemic or rallied 'round and rewarded him, we specify aggregate- and voter-level models assessing the role COVID-19 played in shaping the outcome of the election.[20] Before turning to the voter-level model, we begin by specifying an aggregate-level model assessing whether President Trump gained or lost support in localities hardest hit by the COVID-19 pandemic. To estimate these models, we rely on *The New York Times*' county-level data on COVID-19 severity. This data tracks the cumulative number of confirmed deaths and cases from first confirmed case on January 21, 2020 in Snohomish County, Washington until the day of the election on November 3, 2020.[21] To measure COVID-19 severity in a given county, we specify the overall cumulative deaths and cases, respectively, within a county from the beginning of the pandemic on January 21 as our key variables of interest until Election Day on November 3rd.[22]

We also consider the possibility that voters are sensitive to short-term trends in severity prior to Election Day rather than cumulative severity within their county context. Indeed, some counties experienced COVID-19 more severely (both in terms of cases and deaths) months prior to Election Day. For example, San Diego County in California reported their highest daily increase of COVID-19 deaths on

[19] Five Thirty Eight: How Much Did COVID-19 Affect The 2020 Election? (1/27/2021)https://fivethirtyeight.com/features/how-much-did-covid-19-affect-the-2020-election/

[20] While the results presented in this paper are limited to the presidential election, results for the 2020 US House election are similar. Refer to the supplementary material for House results.

[21] In the appendix, we replicate our forthcoming county-level models at the congressional district level, for which we rely on data provided by the Harvard Center for Population and Development Studies' *COVID-19 Metrics for United States Congressional Districts Project*. This dataset aggregates case and death data to the congressional district level, a salient geographic unit of interest, for the entirety of the COVID-19 pandemic.

[22] Note that we exclude Alaska from the county-level analysis given the lack of consistently defined county-equivalent units across datasets.

September 19th, with 39 new deaths—6 weeks prior to Election Day on November 3rd, 2020. For cases, the county reported its highest daily increase on August 7th, with 652 additional cases—almost three months prior to Election Day. Given gaps like these, we measure long-term COVID-19 severity in terms of cases and deaths by calculating a weekly rolling average throughout our time-series and use the rolling average the week prior to Election Day as our measure of short-term severity. Given stark variation in population across counties, we specify all measures per 10,000 residents.[23] This provides two measures of severity for both cases and deaths that can assess whether cumulative or short-term COVID-19 severity influenced the electoral fortunes of Donald Trump at the top of the ballot. We summarize the measurement of our key independent variables in Table 1.

Since our theory evaluates whether Donald Trump electorally benefited or suffered as a function of local COVID-19 severity, our dependent variable measures the relative swing in Republican support in 2020 relative to the previous presidential election.[24] As such, we construct a measure of the Δ change in the margin between the Republican and Democratic two-party vote shares between the 2020 election (time t) and the previous election in a given office (time $t-1$). Thus, our dependent variable takes the following form:

$$\begin{aligned}&\left[\left(\frac{\text{GOP}\,\%_t}{\text{GOP}\%_t+\text{Dem}\ \ \%_t}\right)-\left(\frac{\text{Dem}\%_t}{\text{GOP}\,\%_t+\text{Dem}\,\%_t}\right)\right]\times 100\\&-\left[\left(\frac{\text{GOP}\,\%_{t-1}}{\text{GOP}\,\%_{t-1}+\text{Dem}\,\%_{t-1}}\right)-\left(\frac{\text{Dem}\,\%_{t-1}}{\text{GOP}\,\%_{t-1}+\text{Dem}\,\%_{t-1}}\right)\right]\times 100\end{aligned}\tag{1}$$

The resulting quantity captures the relative swing in Republican support from the previous presidential election in 2016 to the 2020 election cycle, with positive values articulating a gain in Republican support and negative values articulating a loss in Republican support. For example consider Starr County, Texas; a county situated within the Rio Grande Valley on the Texas-Mexico border that provided the largest relative gain for Donald Trump. The county swung a staggering 56.3% towards Donald Trump from the 2016 to the 2020 election, with Trump losing the county by only a 5.04% margin to Joe Biden, a stark improvement over losing the county by 61.37% to Hillary Clinton in 2016. By contrast, the largest relative loss in support for Trump took place in Henry County, Georgia in suburban Atlanta. Trump lost the county by 20.59% in 2020, a much wider margin than his 4.52% loss in 2016—a swing of 16.17% against President Trump. Figure 1 shows a spatial distribution of this dependent variable in the presidential context. Unsurprisingly, aside from pronounced gains in heavily Hispanic south Texas and Florida, President Trump lost

[23] Except for the measures used in the House analysis presented in the supplementary material that are specified per 100,000 residents. This normalization per 10,000 residents is done by dividing the cumulative number–or rolling seven day average in the case of the short-term severity measure–of COVID-19 cases or deaths (as articulated in Table 1) by the total population of a given county (or congressional district) and multiplying this resulting quotient by 10,000. This allows for direct comparison of COVID-19 case or death severity across counties that vary by population.

[24] For the congressional analyses presented in the supplementary material, the dependent variable is constructed relative to the most recent election for that office.

support in both liberal and conservative counties in his unsuccessful re-election bid against former Vice President Joe Biden, losing an average of 1.11% in relative support in 2020 compared to 2016 across all counties.[25] Indeed, the Trump/Pence ticket performed electorally worse against the Biden/Harris ticket than it did against the Clinton/Kaine ticket in all states with the exception of the District of Columbia, Nevada, Illinois, Arkansas, California, Florida, Utah, and Hawaii.[26] In sum, there is rich variation in the geographic distribution of relative changes in electoral performance for the Trump/Pence ticket from the 2016 to the 2020 election.

Now that we have specified our independent and dependent variables of interest, we turn to the specification of our main county-level empirical model. We specify a total of four models given our measures of COVID-19 severity in both cases and deaths. To assess the impact of deaths on changes in Republican presidential electoral support, we specify two models: the first assesses the effect of long-term cumulative deaths and the second assesses the effect of short-term weekly trend in deaths prior to Election Day. The third and fourth models respectively assess the effects of long- and short-term COVID-19 case rates in the same manner as first and second models. In terms of accounting for other predictors of partisan electoral changes, we build on the standing county-level model assessing electoral changes from one election cycle to the next by Chyzh and Urbatsch (2020). Each model controls for potential constituency predictors of electoral outcomes during the 2020 election, such as the unemployment rate, population density, educational attainment, racial composition of the constituency, median income, constituency lagged Republican vote share, and the party of the governor. The latter two variables seek to control for consistent findings in the literature that: (1) Republican constituencies may be less prone to behavioral modifications (i.e., mask wearing and social distancing) that may contribute to the spread and mortality rate of COVID-19 (Gadarian et al., 2021; Clinton et al., 2021) and (2) Democratic governors, relative to Republican governors, were more likely to mandate public health policies (i.e., stay-at-home and social distancing orders) (Baccini & Brodeur, 2021; Shay, 2020). Indeed, voters may feel compelled to hold a Republican President to greater electoral account if they reside in a constituency overseen by a Republican governor who declines to implement restrictive COVID-19 containment policies or in a strongly Republican constituency where behavioral modifications may be lacking. By accounting for these salient covariates, we better isolate the relationship between COVID-19 cases, deaths, and changes in Republican electoral support.[27]

Lastly, given the hierarchical nature of our county-level data, we specify our models using three distinct statistical frameworks. This is of particular importance given the inherent heterogeneity of COVID-19 across counties—which are nested

[25] To that point, nineteen of the top twenty counties that swung the most towards President Trump are located in south Texas. The one exception was Miami-Dade, Florida, which saw a staggering 22.89% gain in support for Trump.

[26] From 2016 to 2020 Trump performed 0.03% better against Biden than Clinton in DC and NV; while also seeing a 0.08%, 0.70%, 0.95%, 2.16%, 2.40%, 2.72% relative improvement in IL, AR, CA, FL, UT, and HI.

[27] For further details on the coding of these control variables, please visit the manuscript's appendix.

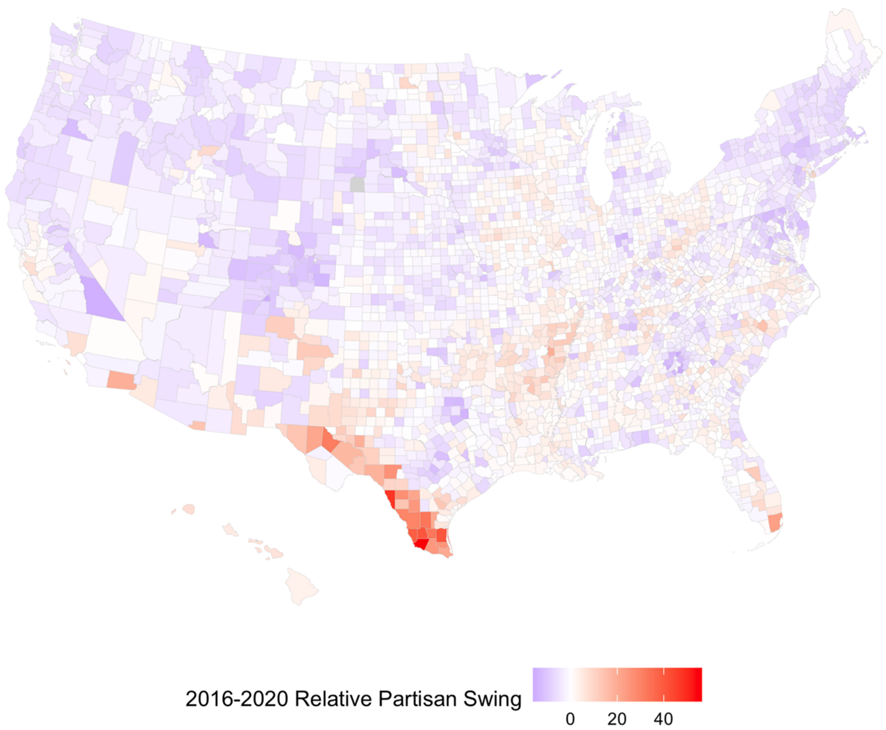

Fig. 1 Δ Change in county-level support for president trump during the 2020 election

Table 1 Summary of key independent variables measuring subnational COVID-19 severity

COVID-19 severity measure	Severity term	Specification formula
Cumulative COVID-19 cases	Long-term severity	$\sum_{1/26/20}^{11/2/20} \text{Cases}$
Weekly trend COVID-19 cases	Short-term severity	$\sum_{10/26/20}^{11/2/20} \text{Cases}/7$
Cumulative COVID-19 deaths	Long-term severity	$\sum_{1/26/20}^{11/2/20} \text{Deaths}$
Weekly trend COVID-19 deaths	Short-term severity	$\sum_{10/26/20}^{11/2/20} \text{Deaths}/7$

within states. To that end, we estimate our models using multilevel ordinary least squares regression with counties (level 1) nested within states (level 2) in a fixed-effects and random-effects framework.[28] We also recognize the likely case that the spatial distribution of COVID-19 cases and deaths may be geographically dependent, with counties more severely affected by COVID-19 to be spatially clustered together. Figure 2 shows the geographic distribution of COVID-19 cases and deaths across counties and provides qualitative support for this notion both in cumulative and short-term weekly trend terms. Across all measures of severity, it is clear that

[28] This baseline state fixed-effects model is specified with state-level clustered standard errors.

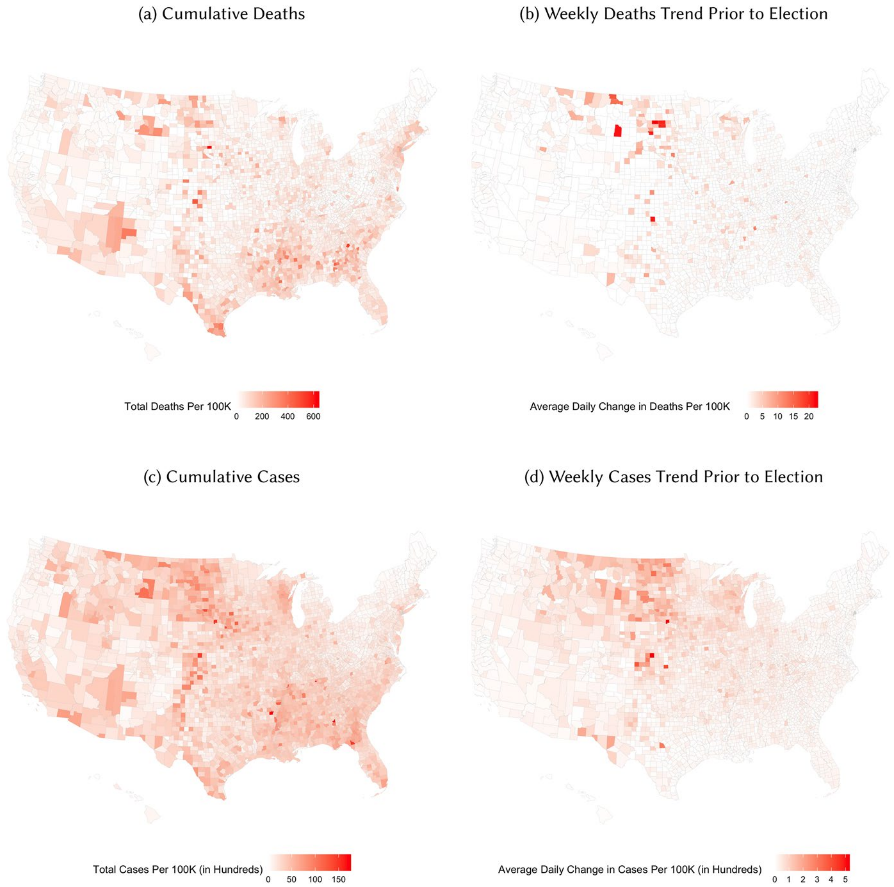

Fig. 2 COVID-19 severity across U.S. counties

rural counties, particularly those in the upper Midwest and deep South, were more impacted by COVID-19 induced deaths and cases. For example, it appears that the trend in weekly cases (i.e., the seven day moving average) prior to the election were disproportionately concentrated in rural counties of the Dakotas, Montana, and Wisconsin. Moreover, spatial dependence exists in the voting patterns of individual counties, with counties exhibiting similar levels of partisan voting as their neighbors (see Darmofal & Strickler, 2019, Ch. 4). Given this concern of spatial dependence, we specify our model using a spatial error framework accounting for this spatial dependence in the error term (see Darmofal, 2015, for a comprehensive description). Subsequent diagnostic checks confirm the presence of spatial dependence and justify the specification of spatial lag and spatial error models .[29]

[29] Specifically, we follow the lead of Darmofal (2015) by employing a set of Lagrange multiplier (LM) diagnostic tests for spatial dependence using our main OLS model. Using the decision rule highlighted by Darmofal (2015), these diagnostic tests both confirm the presence of spatial dependence and indicate

Specifying the Voter-Level Model of Electoral Accountability

We also evaluate our contrasting frameworks assessing how COVID-19 shaped political support for Republican president Donald Trump during the 2020 election using a voter-level model. At the voter level, the presidential rally theory posits that voters should reward President Trump on the basis of their concern for the COVID-19 pandemic while the retrospective accountability framework would hold that voters should punish the Republican president given the pronounced public health costs from the pandemic (Ashworth, 2012; Achen & Bartels, 2016). To that end, recent voter-level work from Mendoza Aviña and Sevi (2021) suggests a degree of retrospective accountability manifested itself in the 2020 presidential election, where voters with close exposure to COVID-19 cases and deaths were less likely to support President Trump in 2020. We build on this model by examining the relationship between support for President Trump in 2020 and individual-level concern about COVID-19 health threat.

This approach presents two clear testable hypotheses with respect to evaluating whether the relationship between the COVID-19 pandemic and political support for President Trump during the 2020 election was subject to a rally or retrospective accountability. We expect that if a rally did emerge at the voter level there should be a positive relationship between concern for COVID-19 as a public health calamity and political support for President Trump. This expectation is consistent with the notion that individual voters rallied behind the President in the face of an exogenous national crisis, in this case a national pandemic (Healy & Malhotra, 2013; Achen & Bartels, 2016).

By contrast, retrospective accountability should manifest in a negative relationship between COVID-19 public health concern and support for President Trump. If voters held President Trump accountable in the 2020 presidential election then those that are concerned with COVID-19 as a public health threat should be less likely to support the president – in line with the findings presented by Mendoza Aviña and Sevi (2021). This negative relationship that theoretically aligns with rational retrospective accountability should be predicated on public perceptions that the Trump administration inadequately responded to the COVID-19 pandemic, at times downplaying the spread of the COVID-19 virus (Baccini & Brodeur, 2021; Fowler et al., 2021; Hart et al., 2020) which was the leading cause of death in the United States in 2020 (Woolf et al., 2021). Indeed, Green et al. (2020) find that early Republican messaging regarding the COVID-19 pandemic centered on downplaying the public health threat and shirking responsibility by placing a greater emphasis on the origins of the virus in China. Given the perceived inadequacy of the Trump

Footnote 29 (continued)
that a spatial error model should be estimated to account for this spatial dependence. In each of our cases and deaths models, and for each electoral context, the robust LM error diagnostic statistic was consistently significant and larger than the robust LM lag diagnostic, justifying our spatial modeling choice. We estimate these models using the `spatialreg` package in `R`. Note that we include specifications of both a spatial lag and spatial error models accounting for geographic dependence towards demonstrating identical and robust substantive results.

Administration's handling of the COVID-19 pandemic[30] and attempts to avoid responsibility, the voters most concerned about COVID-19 as a public health issue should be less likely to support President Trump's 2020 reelection under a model of retrospective accountability.

To evaluate these contrasting empirical expectations at the individual level, we draw on two nationally representative survey waves during the 2020 presidential election provided by the Democracy Fund Voter Study Group Panel (VOTER) Survey and conducted by YouGov. The two survey waves, conducted prior to the election in September and again after the election in November, measure four (4) key outcome variables with respect to political support for President Trump.[31] The two waves of survey data allow us to measure political support for President Trump by asking respondents: (1) if they approved of President Trump's handling of the COVID-19 pandemic and (2) if they preferred President Trump over Former Vice President Joe Biden in the 2020 election. Moreover, given the panel nature of the VOTER survey with respondents being asked which candidate they supported in the 2016 election, we can also measure political support for President Trump in the form of: (3) switching their electoral preference from Democrat Hillary Clinton in 2016 to Republican Donald Trump in 2020 and (4) switching their electoral preference from Republican Donald Trump in 2016 to Democrat Joe Biden in 2020. To evaluate these four outcome variables for each of the two survey waves, we specify a logistic regression model predicting political support for President Trump as a function of concern for COVID-19 as a public health threat and a vector of control variables, such as partisanship, local COVID-19 severity context, education, race, gender, ideology, retrospective economic evaluations, personal proximity to a COVID-19 infection (i.e., having contracted COVID or knowing someone who contracted it), and age.[32] Our key independent variable is a simple binary measure, coded 1 if a respondent indicated that they were somewhat or very concerned "that you or a close family member will get sick from the coronavirus" or 0 if a respondent indicated that they were not at all or not very concerned. In addition to our baseline additive expectation that public health concern will decrease political support for President Trump, we specify multiplicative models interacting public health concern

[30] Reuters: Trump's handling of coronavirus pandemic hits record low approval: Reuters/Ipsos poll (10/8/2020)https://www.reuters.com/article/us-usa-election-trump-coronavirus/trumps-handling-of-coronavirus-pandemic-hits-record-low-approval-reuters-ipsos-poll-idUSKBN26T3OF

[31] Specifically, the September VOTER survey wave conducted interviews between August 28th and September 28th, 2020 while the November interviews were conducted between November 13th and December 7th, 2020.

[32] For a comprehensive description of model covariates, please see the appendix. Given that the lowest geographic context unit is at the congressional district level, we measure local COVID-19 severity as the cumulative COVID-19 death rate in a respondent's congressional district at the time of their survey completion. We rely on data provided by the Harvard Center for Population and Development Studies' *COVID-19 Metrics for United States Congressional Districts Project* that measures COVID-19 deaths at the congressional district-level. We also leverage this dataset to evaluate our county-level models within the context of congressional elections in the appendix, which provides for congruent results.

with partisanship to evaluate whether the relationship between public health support and support for President Trump differs by partisanship (Heersink et al., 2020).[33]

Results: COVID-19 & President Trump's Performance

Now that we have specified both aggregate-level and voter-level models evaluating whether the the rally or retrospective accountability frameworks best describes the relationship between COVID-19 severity and President Trump's 2020 electoral performance, we turn to evaluating the models.

We begin by evaluating the aggregate model shown in Fig. 3, which assesses the relationship between county-level COVID-19 severity and change in relative electoral support for Republican President Donald Trump in the 2020 election.[34] Note that Fig. 3 and all forthcoming figures display point estimates with corresponding 90% and 95% confidence intervals. Each panel of Fig. 3 assesses the relationship between a given county-level measure of COVID-19 severity (cases, deaths) across four differing statistical estimator specifications: fixed-effects, random-effects, spatial error, spatial lag. Figure 3 Panels A and B assess the cumulative severity of COVID-19 cases and deaths on the county-level change of electoral support for Donald Trump from 2016 to 2020. As one can see in Fig. 3 Panel A, the cumulative number of COVID-19 cases in a given county was a largely significant predictor of changes in relative electoral support for Donald Trump. With the exception of the fixed-effects model, all models in Fig. 3 Panel A report a positive significant relationship between the cumulative COVID-19 cases at the county level and relative support for Donald Trump. Figure 3 Panel B finds significant evidence that more COVID-19 deaths are correlated with greater relative electoral gains for Donald Trump from 2016 to 2020 across all model specifications. The size of the coefficient of cumulative COVID-19 deaths range from 0.041 in the spatial error estimation to 0.072 in the fixed-effects, and random-effects, estimation. For substantive interpretation of this positive significant effect, consider the counties of Powatan (VA), Moultrie (IL), and Foard (TX), which represent the lower, median, and upper quartiles of counties in terms of cumulative COVID-19 deaths per 100,000 residents, respectively. Taking the results of the fixed-effects models, President Trump modestly gained approximately 0.13% (1.74×0.072), 0.30% (4.10×0.072), and 0.56% (7.84×0.072) over his 2016 electoral performance in these three counties

[33] Given the nested nature of the survey data, we specify our individual-level logistic regression models with congressional district-clustered robust standard errors.

[34] In the appendix, we follow the guidance of Lenz and Sahn (2020) and report bivariate relationships, along with specification plots, of our effects of interest to account for control variable-induced increases in estimated effect sizes and statistical significance. Across all specifications, our significant results hold and are thus *not sensitive* to the inclusion of various control variables of theoretical importance. We also report results from a sensitivity analysis articulating the robustness of our key positive relationships of interest in the face of potential omitted variable bias (Cinelli & Hazlett, 2020). Note that our models are identical when switching the outcome variable from 2016-2020 Republican gain to 2020 two-party Republican vote-share. These results can also be found in the appendix.

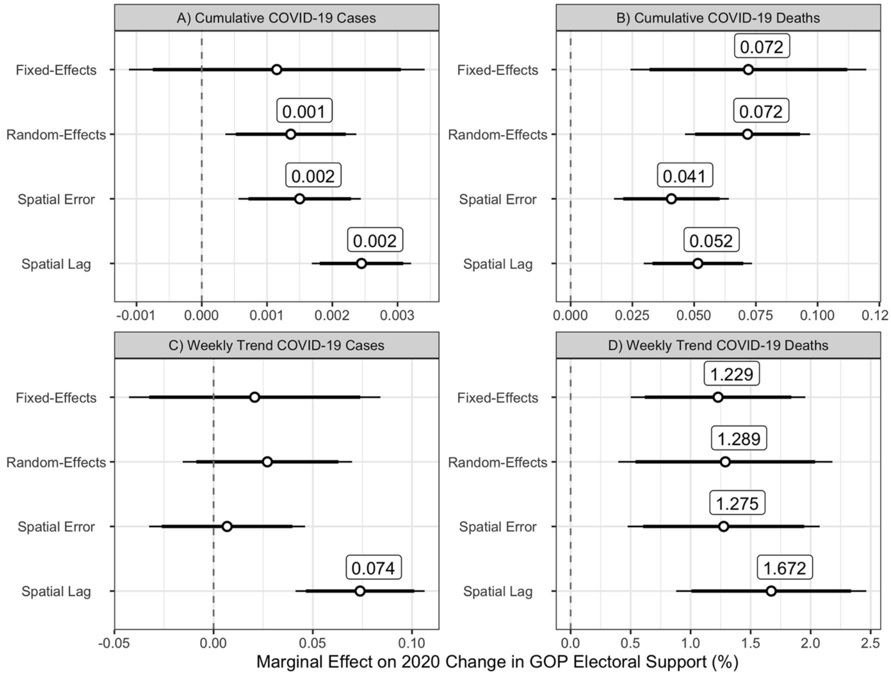

Fig. 3 County COVID-19 severity & change in GOP presidential electoral support. Estimates articulating the relationship between county COVID-19 severity and change in electoral support for President Trump in Fig. 3 are derived from full models specified with demographic and partisan control covariates. Coefficients significant at the 95% confidence level labelled

after accounting for traditional predictors of electoral outcomes.[35]. As such, a one standard deviation in increase in cumulative county-level COVID-19 deaths, such as going from Candler (GA) to Starr (TX), is correlated with a 0.25% (6.04×0.041) increase in Trump support in the fixed-effects estimate.

Turning to short-term case severity measure evaluated in Fig. 3 Panel C, we find a lack of evidence that the severity of COVID-19 cases, prior to election day, in a given county correlated with changes in electoral support for Donald Trump. However, Fig. 3 Panel D finds evidence of a significant relationship between short-term severity of COVID-19 deaths, measured as the weekly seven-day Δ trend in

[35] Note that this interpretation applies to the fixed-effects and random-effects model coefficients presented in Fig. 3 Panel B. The spatial error model coefficients, which can be interpreted just like OLS coefficients, would yield similar results by showing that Donald Trump gained approximately 0.07% (1.74×0.041), 0.17% (4.10×0.041), and 0.32% (7.84×0.041) in these three counties. Interpretation of the random-effects models are identical to the fixed-effects models.

a county's death rate prior to the election, and an increase in electoral support for Donald Trump across all statistical estimators. Since most counties experienced no significant changes in COVID-19 mortality trends the week prior to the election ($\eta \approx 0$), consider the 60%, 80%, and maximum quantile county cases of Seminole (FL), Clay (MS), and Logan (ND). Again, taking the results of the fixed-effects models, President Trump gained approximately 0.02% (0.02×1.229), 0.09% (0.07 ×1.229), and 2.81% (2.29×1.229) over his 2016 relative electoral performance in these three counties that were experiencing a rapid short-term Δ increase in COVID-19 deaths the week prior to Election Day. Taken together, it is clear that Donald Trump gained on the basis of COVID-19 death severity in long-term cumulative and short-term trend terms. While these significant gains may appear modest, it is important to note that these county-level substantive gains due to COVID-19 deaths are roughly on par with President Trump's losing statewide margins in Arizona (0.31%), Georgia (0.24%), and the tipping-point state of Wisconsin (0.63%). After accounting for all standard predictors of Donald Trump's county-level performance, it appears that his electoral gains from COVID-19 death severity may have kept the statewide margin in these critical battleground states close.

By contrast, results of the individual-level model shown in Fig. 4 point to a lack of a COVID-19 rally for President Trump, with some evidence that public health concern for COVID-19 correlates with a decline in Republican support. As shown in Fig. 4, the additive baseline relationship between public health concern for COVID-19 and support for President Trump is significant and negative across both the September and November survey panels in models predicting COVID-19 job approval and a Republican vote in the presidential election. Indeed, agreeing that COVID-19 poses a public heath threat to either oneself or one's family is associated with a decline in the probability of approving of President Trump's handling of COVID-19—or stating an electoral preference for Trump—by about 4% and 7% in the September and November panel waves, respectively. Results of the approval and electoral preference models by partisanship also show a general consistency in this negative relationship between public health concern and political support for President Trump, with Republican, independent, and Democratic partisans being less likely to give pandemic-handling approval or espouse electoral support for the Republican president on the basis of public health concern.

While public health concern is not predictive of the very small number of voter cases in the survey data that reported switching the partisan preference of their votes from the 2016 to the 2020 elections (i.e., switching from Clinton 2016 to Trump 2020 and vice-versa), the general findings of the individual-level model provides a noticeable contradiction of the aggregate model.[36] This contradiction is rooted in our observation that while the county-level models find robust evidence that President

[36] Reflecting the remarkable consistency in partisan electoral preferences–and rare event of partisan defection—from one election to the next, we note that across both the September and October waves of the VOTER survey only 32/3,180 (45/3,180) and 24/2,872 (49/2,872) reported a 2016-2020 switch in partisan presidential preference from Democratic to Republican (Republican to Democratic). Indeed, party switchers represent about 1% of all survey respondents across both survey waves.

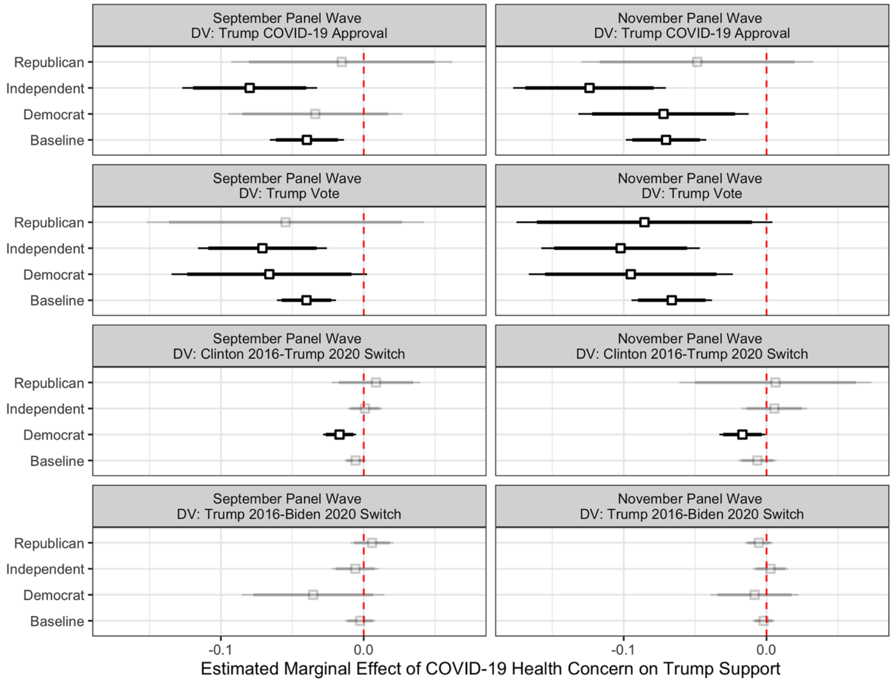

Fig. 4 Relationship between public health concern & political support for president Trump. The discrete marginal effects articulated in Fig. 4 specified with 90% and 95% confidence intervals from district-clustered robust standard errors. Darker shaded point estimates significant at $\rho < 0.10$. Models control for local COVID-19 context, education, race, gender, ideology, economic evaluations, COVID-19 infection proximity, and age. Δ change in electoral choice models also control for change in partisanship and 2016 Republican presidential vote. Fig. 4 articulates N = 8 unified baseline additive models (one for each panel wave-outcome variable interest articulated in the panel) 8 unified interactive models (one for each panel wave-outcome variable interest articulated in the panel) for a total of N = 16 models. Heterogeneous effects of attitudes across partisanship estimated from multiplicative model interacting context and partisanship. All models fitted with wave-specific survey weights

Trump gained in areas of high COVID-19 severity, the individual-level results show clear evidence that President Trump did not experience a rally effect and even paid an electoral price among voters that viewed the pandemic as a public health threat.

In the forthcoming section, we aim to reconcile these seemingly contradictory results by proposing and testing an alternative theory that argues that local COVID-19 severity context correlates with economic concern regarding state COVID-19 restrictions, which in turn served as a catalyst for political support for President Trump. This argument also allows us to posit that the preceding voter-level model is incomplete since it lacks a measure of voter attitudes regarding COVID-19 restrictions. Incorporating these attitudes provides our individual-level models with more validity, given that concerns about COVID-19 manifested themselves in both the public health and economic dimensions rather than exclusively as a public health threat (Neundorf & Pardos-Prado, 2021). We test this revised framework and find robust evidence for the proposed voter-level mechanism that President Trump's gains due to COVID-19 were enabled by his activation of voters' economic concerns centered around restrictions.

Reconciling the Puzzle: Considering Context, COVID-19 Restriction Attitudes & GOP Support

Context & COVID-19 Attitudes

To test the argument that local COVID-19 context motivates concerns about restrictions and public health concerns in differing ways, we return to the two VOTER survey panel waves. We specify two baseline additive models with one outcome variable measuring COVID-19 public health concern and the other COVID-19 restrictions concern. We measure the outcome variable of COVID-19 public health concern as previously described, coded 1 if a respondent indicated that they were somewhat or very concerned "that you or a close family member will get sick from the coronavirus" or 0 if a respondent indicated that they were not at all or not very concerned. To measure economic concerns surrounding state-imposed COVID-19 restrictions, we turn to the survey question asking respondents "thinking about the decisions by a number of state governments to impose significant restrictions on public activity because of the coronavirus outbreak, is your greater concern that state governments will either (1) 'not lift the restrictions quickly enough' or (2) ''lift the restrictions too quickly." We code this measure of COVID-19 restrictions concern consistent with our theoretical expectations that President Trump electorally benefited from restriction concerns (and primed voters to these in his campaign), we code this restrictions outcome variable (1) if a respondent stated that they believed restrictions would not be lifted quickly enough and (0) if the response was that restrictions would be lifted too quickly. The logic behind this analytical model is consistent with our theoretical argument, that those living in localities with high COVID-19 severity would be more anxious about the possibility that COVID-19 restrictions would not be lifted quickly enough due to associated negative economic ramifications rather than those in low COVID-19 severity localities. As such, we expect there to be a positive relationship between local COVID-19 severity and our outcome variable of restriction concerns. To explore potential partisan heterogeneity in the relationship between severity context and our two outcome variables, we interact severity with partisanship in each of our survey panel wave models.[37]

Figure 5 articulates our model results assessing the relationship between COVID-19 restriction severity and our two outcome variables of interest across each survey wave. As one can see in the top two panels of Fig. 5 assessing the results of our public health concerns models, our additive baseline results show a null relationship between COVID-19 severity context and the probability of viewing COVID-19

[37] Moreover, we also specify each individual-level model in a manner consistent with the models in the previous section, with a standard vector of control variables, district-clustered standard errors, and specification with wave-specific survey weights.

as a public heath concern threat across both the September and November surveys. This null effect of severity generally holds across partisan groups in both survey waves in the multiplicative models, underscoring that all partisans held consistent beliefs regarding COVID-19 as a public health concern independent of local context.[38] Interestingly, results of the COVID-19 health concern models show that, in only the September wave, that Republicans and independents were less likely to view COVID-19 as a public heath concern than Democrats by about 7% and 4%, respectively (in line with previous research by Craig et al. 2022; de Bruin et al. 2020). Taken together, our individual-level models show that context is an insignificant predictor of COVID-19 public health concerns.

By contrast, the results of our restrictions concern models in the bottom two panels of Fig. 5 provide support for our proposition that local severity context predicts these restriction attitudes. The additive model shows that local COVID-19 death severity is predictive of a voter being concerned that state-imposed COVID-19 restrictions would not be lifted quickly enough as opposed to being lifted too quickly. Indeed, going from the minimum to maximum value of local COVID-19 death severity coincides with a predicted increase in restrictions concern by 14.3% and 17.9% in the September and November survey wave models, respectively. This significant contextual relationship is similar in magnitude to the relationship between partisanship and restriction concerns. Relative to Democratic partisans, independents and Republicans were 13.0% (14.8%) and 15.8% (22.2%) more likely to espouse a concern that restrictions will not be lifted quickly enough as opposed to too quickly in the September (November) panel survey wave models. These additive baseline findings reveal that, in addition to partisanship, severity context predicts COVID-19 restriction attitudes across both survey waves.

Turning to partisan heterogeneity estimated from the interactive model framework in the bottom two panels of Fig. 5, we find consistent evidence across both survey waves that the relationship between context and probability of restrictions concerns varies among Democrats and not Republicans. Perhaps this reflects the general agreement among Republicans regarding COVID-19 restrictions, which was a salient campaign framing from President Trump. Indeed, our multiplicative model finds no heterogeneity in the probability of espousing a restriction concern among Republicans on the basis of varying local severity COVID-19 context. By contrast, we find evidence of heterogeneity in the probability of Democrats voicing a restriction concern on the basis of local COVID-19 severity across both survey waves. Going from the minimum to maximum value of local COVID-19 death severity coincides with a predicted increase in restrictions concern by 23.2% and 24.9% among Democratic respondents in the September and November wave, respectively. In contrast to Republicans, Democrats did react to local COVID-19 severity when shaping their views on COVID-19 restrictions, with Democrats in harder hit locality being more concerned that restrictions would not be lifted quickly enough (relative

[38] Indeed the key exception to this is the finding that among Republican partisans in both waves, with greater local COVID-19 severity correlating with greater likelihood of perceiving COVID-19 as a public health concern.

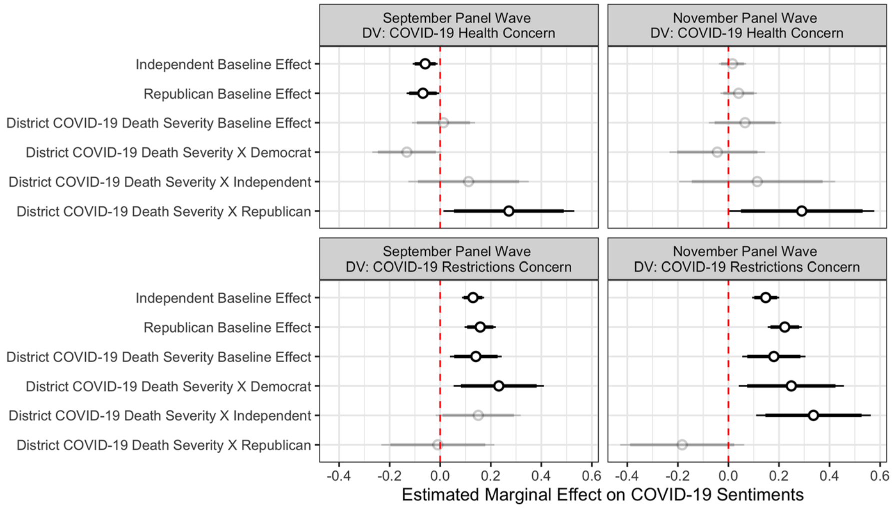

Fig. 5 Relationship Between Local Severity Context & COVID-19 Attitudes. The minimum-maximum first difference marginal effects articulated in Fig. 5 specified with 90% and 95% confidence intervals from district-clustered robust standard errors. Darker shaded point estimates significant at $\rho < 0.10$. Models control for education, race, gender, ideology, economic evaluations, COVID-19 infection proximity, and age. Figure articulates N = 4 unified baseline additive models (two per panel wave x two outcome variables of interest) 4 unified interactive models (two per panel wave x two outcome variables of interest) for a total of N = 8 models. Heterogeneous effects of local context on attitudes estimated from multiplicative model interacting context and partisanship. All models fitted with wave-specific survey weights

to lifting restrictions too quickly) than their co-partisans in lower severity context. Taken together, this analysis finds strong support that local COVID-19 severity context does shape attitudes regarding COVID-19 restrictions in a manner that does not manifest itself when modelling concern for COVID-19 as a public health threat.

Restriction Attitudes & Increased GOP Support

As previously argued at the voter-level, we posit that local COVID-19 context predicts attitudes about state-imposed COVID-19 which, in turn, predicts greater likelihood of supporting President Trump in the 2020 election. Given that we find support that COVID-19 context motivates restriction attitudes, we turn to the second stage of our revised mechanism framework, where we expect that restriction attitudes may have been a catalyst for supporting President Trump in the 2020 elections. In contrast to public health attitudes, we posit that beliefs that restrictions would not be lifted quickly enough would be positively correlated with support for President Trump in the 2020 election. To test this this second stage of our proposed mechanism, we respecify our initial voter-level models articulated in Fig. 4 predicting support for President Trump during the 2020 election to include not only public health concerns but also restriction attitudes as a predictor. We keep all model specification

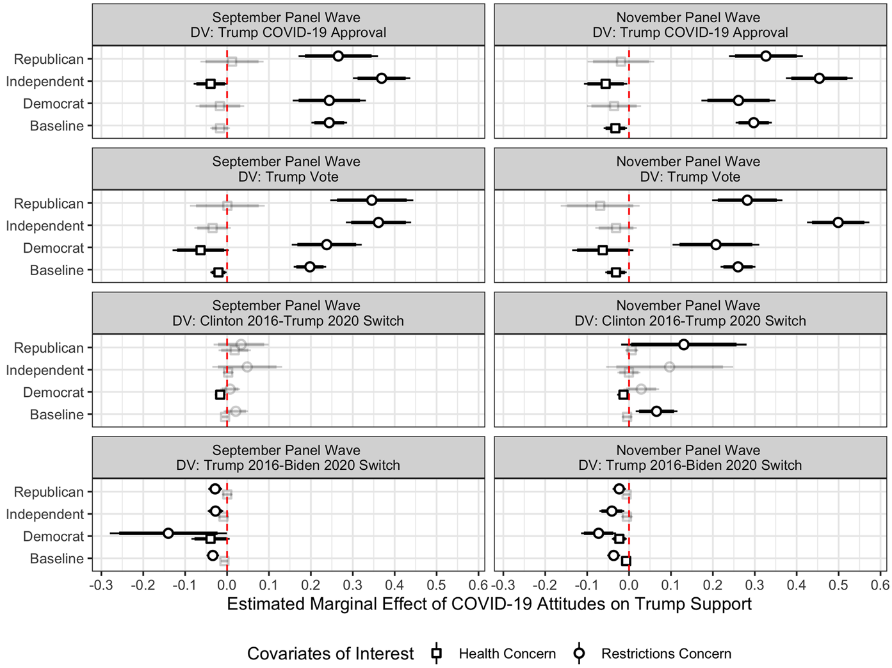

Fig. 6 Relationship between public health concern/restrictions concern & political support for president Trump. The discrete marginal effects articulated in Fig. 6 specified with 90% and 95% confidence intervals from district-clustered robust standard errors. Darker shaded point estimates significant at $\rho < 0.10$. Models control for local COVID-19 context, education, race, gender, ideology, economic evaluations, COVID-19 infection proximity, and age. Δ change in electoral choice models also control for change in partisanship and 2016 Republican presidential vote. Figure articulates N = 8 unified baseline additive models (one for each panel wave-outcome variable interest articulated in the panel) 8 unified interactive models (one for each panel wave-outcome variable interest articulated in the panel) for a total of N = 16 models. Heterogeneous effects of attitudes across partisanship estimated from multiplicative model interacting context and partisanship

attributes consistent with the previous models and simply respecify the models to account for restriction attitudes as a salient predictor.

Figure 6 presents the results of our more comprehensive voter-level models predicting support for President Trump across four different outcome variables and two VOTER survey waves.[39] We find consistent baseline support that COVID-19 restriction concern attitudes correlates with political support for President Trump across both survey waves and all measured outcome variables. In the September (November) panel wave additive baseline models, we find that a concern that COVID-19 restrictions will not be lifted quickly enough significantly correlates with an increase in the likelihood of: (1) approving of President Trump's handling of the COVID by 24.3% (29.7%) and (2) stating an electoral preference for President Trump by 19.8%

[39] We present full model results in the appendix and also replicate the same substantive findings of the models with an outcome variable indicating Republican preference in the 2020 U.S. House elections.

(26.0%) in the September (November) wave panel models. Moreover, we also find that a concern that COVID-19 restrictions will not be lifted quickly enough significantly correlates with an increase in likelihood of: (3) rare vote switching from supporting Democrat Hillary Clinton in 2016 to supporting President Trump in 2020 by 6.6%. Further evaluating rare vote switching from the 2016 to 2020 election cycles, we also find in the September (November) wave panel models that restriction concerns also correlate with a decline in the likelihood of: (4) switching from supporting Republican Trump in 2016 to supporting Democrat Biden in 2020 by 3.4% (3.7%).[40] The multiplicative models show that the relationship between restriction attitudes and political support does not vary substantively across each partisan group, providing evidence that President Trump was able to cultivate political support in a similar manner across Republican, independent, and Democratic partisans on the basis of COVID-19 restriction attitudes. Importantly, including restriction attitudes in the models largely negates the salience of public health attitudes as a predictor of political support for President Trump.

As Fig. 6 clearly shows, in the models were public health concern attitudes are significant negative predictors of political support for President Trump—such as in both waves of the electoral preference models—this relationship pales in comparison to the relationship between COVID-19 restriction attitudes and political support for President Trump. Taken together, the results of our revised comprehensive individual-level models show that COVID-19 attitudes relating to concerns that restrictions would be implemented for too long were more salient predictors of 2020 electoral support than concerns of COVID-19 as a public health threat—a dynamic that benefited President Trump. Taken in conjunction with the results shown in Fig. 5, we identify a potential mechanism that explains our aggregate findings that President Trump gained in areas hardest hit by COVID-19 deaths. Voters in these counties were more likely to be concerned that these economically costly restrictions would not be lifted quickly enough, which in turn served as a catalyst of support for President Trump's re-election efforts. As such, we find empirical support for the expectation that President Trump's campaign appeals to activate attitudes regarding the economic ramifications of the COVID-19 restrictions proved to be an electoral viable strategy in the face of a challenging electoral landscape.

Discussion: The Public Health & Economic Facets of COVID-19

From the reporting of the first case of COVID-19 on January 26th in Snohomish County, Washington to Election Day on November 3rd, the United States experienced over 9,400,000 cases and 232,000 COVID-19 induced deaths. In this paper, we assess whether the pronounced COVID-19 pandemic affected voter behavior and

[40] While not presented in the figure, we find that the relationship between COVID-19 severity context and political support for President Trump to be null across all models, providing suggestive evidence that the individual-level relationship between severity and political support is mediated through COVID-19 restriction attitudes. See the appendix for additional models.

attitudes during the 2020 US presidential election. The two relevant prevailing and competing theories of public political behavior put forth in the literature are 1) the public rallies around the president during exogenous shocks (Baum, 2002; Hetherington & Nelson, 2003) and 2) the public punishes the president retrospectively for mishandling these shocks (Fiorina, 1978; Ferejohn, 1986). Prior studies on the 2020 presidential election provide mixed results – while some report a negative relationship between COVID-19 severity and Trump vote share (e.g., Baccini, Brodeur & Weymouth, 2021) others find that Trump gained electorally in areas where COVID-19 severity was more pronounced. We reconcile these findings by incorporating economic concerns related to the COVID-19 restrictions (e.g. lockdowns) into our model.

In the months leading up the 2020 presidential elections, voters became increasingly concerned about the economic fallout from COVID-19 related restrictions. A joint Morning Consult and Politico poll found that in April of 2020 that 35% of voters were more worried about COVID-19's economic impact than its public health impact—a 6% increase from the previous month.[41] At the elite level, Republican officials focused their messaging on the economic costs of pandemic restrictions rather than public health concerns. In the months leading to the 2020 election, Trump attacked Biden over pandemic restrictions, stating that "[Biden] wants to shut down this country, and I want to keep it open."[42]

In this study, we argue that concerns about COVID-19 related economic restrictions independently influenced vote choice in the 2020 presidential election. At the county level, we show that higher COVID-19 severity was associated with greater Trump support. Using voter-level data we show that greater concern about the economic impacts of COVID-19 restrictions was associated with increased support for Trump in 2020 while concern about public health was either uncorrelated or very weakly correlated with decreased support for Trump. While our results reveal that the individual-level magnitude of the effect of economic concerns on vote choice was greater than that of public health concerns, this was not enough to swing the election to Trump, as approximately 60% of voters nationwide aligned were more concerned about restrictions being lifted too quickly—in alignment with a public health framework. Indeed, the crosstabs reported in Table 2 support these contentions. While President Trump gained among some members of the public by priming economic effects, these primes were ultimately not fruitful as only 40% of the public across both survey waves were concerned that restrictions would not be lifted quickly enough. Moreover, the proposition that restrictions were being lifted too slowly enjoyed substantially less than majority support among independent partisans, a key voting bloc during President Trump's ultimately unsuccessful 2020 re-election bid.

[41] Morning Consult: COVID-19's Economic Impact Starts to Take Center Stage for GOP Voters (04/22/2020)https://morningconsult.com/2020/04/22/coronavirus-economy-public-health-concern-gop/

[42] CNN: Does Biden want to shut down the country?https://www.cnn.com/factsfirst/politics/factcheck_80d8fd55-b101-4dcc-8a84-2a43582f6832

Table 2 Percent of Partisans Who Believe COVID-19 Restrictions Are Being Lifted Too Slowly

	September panel wave	November panel wave
General public	38.47 (0.72)	42.22 (0.87)
Democrat	14.32 (0.94)	15.04 (1.04)
Independent	41.16 (0.94)	45.20 (1.02)
Republican	68.75 (1.11)	73.59 (1.19)

Estimates are weighted mean percentages with standard errors in parenthesis.

September Wave: Public $N = 5900$; Democrats $N = 2270$; Independents $N = 2171$; Republicans $N = 1459$.

November Wave: Public $N = 4943$; Democrats $N = 1935$; Independents $N = 1792$; Republican $N = 1216$

Beyond national electoral effects, our study provides greater insight into how local experiences of the pandemic affected vote choice and attitudes about COVID-19. We show that local COVID-19 severity is associated with greater concern about the economic impacts of restrictions, but not with greater concern about the public health dimension of the pandemic. These findings suggest an voter-level mechanism that explains disagreements in prior studies of the 2020 election and helps explain why Trump gained electorally in areas where COVID-19 was more pronounced. Higher local levels of COVID-19 deaths were associated with greater voter concern about economic costs (which electorally assisted Trump) but not associated with concerns about public health (which provided at most a mild electoral boost for Biden). This finding also demonstrates the difficulty of applying either the rally effect or retrospective accountability theory as a one-size-fits-all model to explain voter reactions to a national crisis.

In addition to these contributions, this paper offers several additional avenues that merit further study. While our research demonstrates clear and novel relationships between pandemic attitudes, local COVID-19 severity, and presidential vote choice, there are many questions that remain. As we show (see Fig. 5), the relationship between local COVID-19 severity and individual-level concerns about both COVID-19 health risks and COVID-19-related restrictions is conditioned by partisanship. While Democrats and independents were more likely to to be concerned about restrictions if they lived in a county with high COVID-19 death rates, the same is not true for Republicans. Conversely, our results show that while Republicans were more likely to to be concerned about public health if they lived in a county with high COVID-19 death rates, the same is not true for Democrats and independents. These results are perhaps suggestive of a cross-pressuring dynamic, whereby partisans are updating their prior beliefs in response to local exposure. But such a mechanism is only suggested here, and should be explicitly explored in future research. We hope that the results we have presented here provide scholars with a springboard to assess the electoral implications of the COVID-19 pandemic during the 2020 presidential election.

While this paper provides new insights, additional research is still needed on the role of COVID-19 in the 2020 presidential election. We have proposed that opposition to restrictions fueled Trump support in areas with high COVID-19 severity, and provide suggestive evidence for this theory. We encourage future researchers to interrogate this proposed mechanism and test it against alternative explanations. For instance, rather than Trump's support growing as a backlash against restrictions in areas of high COVID-19 severity, it is possible that the causal arrow runs in the opposite direction. Perhaps areas with high Trump support already agreed with his sanguine view of the pandemic, prioritized the economy over public health, and were less likely to comply with COVID-19 restrictions—leading to higher levels of COVID-19 severity. In such a scenario, Trump is not necessarily a beneficiary of COVID-19 severity. While we believe that the mechanism proposed in the body of this paper is compelling (especially given that Democratic partisans with high restriction concerns were more likely to support Trump in the 2020 election), there are any number of alternative explanations worth investigating to explain the findings presented here.

Normatively, this study raises important questions about presidential accountability during national crises. Voters nationwide did punish Trump for his perceived mishandling of COVID-19 – in line with popular media narratives during and following the election. However the effects of COVID-19 at both the individual and aggregate levels are much more dynamic. The counties worst hit by the pandemic were more likely to support Trump, and our results suggest that the cause was a county-level increase in the salience of economic concerns related to pandemic restrictions—which Trump and his co-partisans highlighted in the 2020 campaign. Much as Achen and Bartels (2016) have called into question the "folk theory of democracy," our study challenges classic models of how voters reward (via a rally) or punish (retrospectively) politicians for their handling of a crisis. While many observers assumed that widely reported public dissatisfaction with Trump's handling of the COVID-19 pandemic meant that he would be severely punished by voters in 2020, our results suggest that caution and skepticism are merited in the face of such predictions. Such skepticism is especially merited in light of the divergence in attitudes between partisan groups in our individual-level results. As seminal works like those be Lenz (2013) and Achen and Bartels (2016) have argued, it is likely (particularly in our contemporary political environment) that voters' political attitudes are shaped by co-partisan elites and social group identities, and we should exercise caution when predicting whether voters will reward or punish politicians in line with classical models of democratic accountability.

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